Assignment 2

CSE 423

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Section : 10

Ans to the Qs no-1

$x_{min} = -30$

$x_{max} = 30$

$y_{min} = -30$

$y_{max} = 30$

$x_1 = 35$, $y_1 = 37$, $x_2 = -38$, $y_2 = -20$

For $(35, 37)$

Outcode 1: 1010

For $(-38, -20)$

Outcode 2: 0001

Outcode 1 AND Outcode 2: 0000

[Partially inside]

Outcode 1 has right bit

Applying right boundary intersection:

$x = x_{max} = 30$

$y = y_1 + m(x_{max} - x_1) = 37 + \frac{-20-37}{-38-35}(30-35) = 33.096$

Outcode 1: 1000 (recalculated)

Outcode 1 AND Outcode 2 = 0000

$x_1 = 30$, $y_1 = 33.096$, $x_2 = -38$, $y_2 = -20$

Outcode 1 has top bit,

$y = y_{max} = 30$

$x = x_1 + \frac{1}{m}(y_{max} - y_1) = 35 + \frac{-38-30}{-20-33.096}(30-33.096) = 31$

Outcode 1: 0001

Outcode 1 AND outcode 2 = 0001

So completely outside

Ans to the Q: no-3

t_E	t_L	Comments
0.3	0.7	Accepted
-0.3	1.2	Accepted
-0.3	-0.33	Rejected
-0.5	0.5	Accepted
1.1	2.4	Accepted
0.3	1.3	Accepted